

Day 4: Introduction to OSPF (open shortest path first)

Task 1: Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

Three Cisco 2620XM routers (R1, R2, R3) were placed in Packet Tracer and connected using serial (DCE/DTE) links, forming a closed triangle so that every router has a direct link to the other two. This design gives OSPF two possible paths between any pair of routers.

Each router also has a LAN-side network representing a separate department, and each of these LANs was placed into its own OSPF area (Areas 1, 2, and 3) while the three serial links between the routers form the backbone, Area 0.

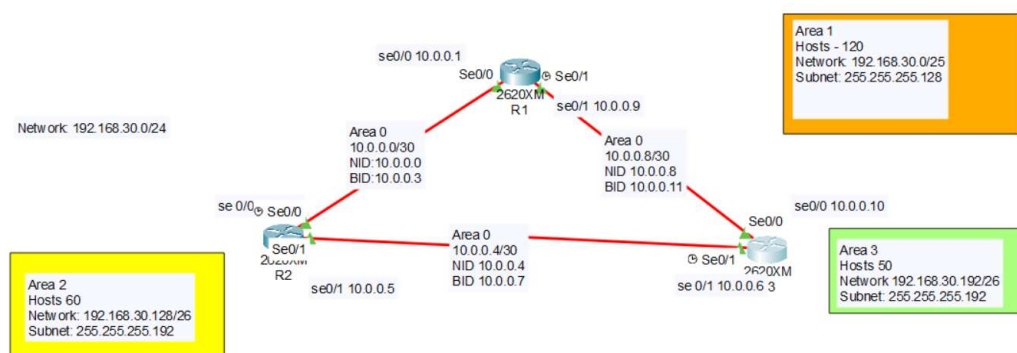


Figure 1 — Triangle topology: R1, R2, and R3 connected via serial links (Area 0 backbone), each attached to its own stub area (1, 2, 3).

Task 2: Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

The three backbone links were each addressed with a /30 subnet (only 2 usable hosts needed per point-to-point serial link), and the three LAN networks were sized to match the host-count requirements given for each area using VLSM.

Link / Network	Router : Interface	IP Address	Area
10.0.0.0/30	R1 : Se0/0	10.0.0.1	0 (backbone)
10.0.0.0/30	R2 : Se0/0	10.0.0.2	0 (backbone)
10.0.0.4/30	R2 : Se0/1	10.0.0.5	0 (backbone)
10.0.0.4/30	R3 : Se0/1	10.0.0.6	0 (backbone)
10.0.0.8/30	R1 : Se0/1	10.0.0.9	0 (backbone)
10.0.0.8/30	R3 : Se0/0	10.0.0.10	0 (backbone)
192.168.30.0/25	R1 : LAN (120 hosts)	192.168.30.1	1
192.168.30.128/26	R2 : LAN (60 hosts)	192.168.30.129	2
192.168.30.192/26	R3 : LAN (50 hosts)	192.168.30.193	3

Task 3: Configure OSPF on each router using the `router ospf 1` command and the `network` command to advertise all directly connected networks.

OSPF process 1 was enabled on all three routers. Each network statement uses a wildcard mask (the inverse of the subnet mask) and explicitly assigns the network to the correct area — the backbone links to Area 0, and each router's own LAN to its dedicated stub area. This makes R1, R2, and R3 each an ABR (Area Border Router) for their respective non-backbone area.

```
R1(config)# router ospf 1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 10.0.0.8 0.0.0.3 area 0
R1(config-router)# network 192.168.30.0 0.0.0.127 area 1
```

```
R2(config)# router ospf 1
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 10.0.0.4 0.0.0.3 area 0
R2(config-router)# network 192.168.30.128 0.0.0.63 area 2
```

```
R3(config)# router ospf 1
R3(config-router)# network 10.0.0.4 0.0.0.3 area 0
R3(config-router)# network 10.0.0.8 0.0.0.3 area 0
R3(config-router)# network 192.168.30.192 0.0.0.63 area 3
```

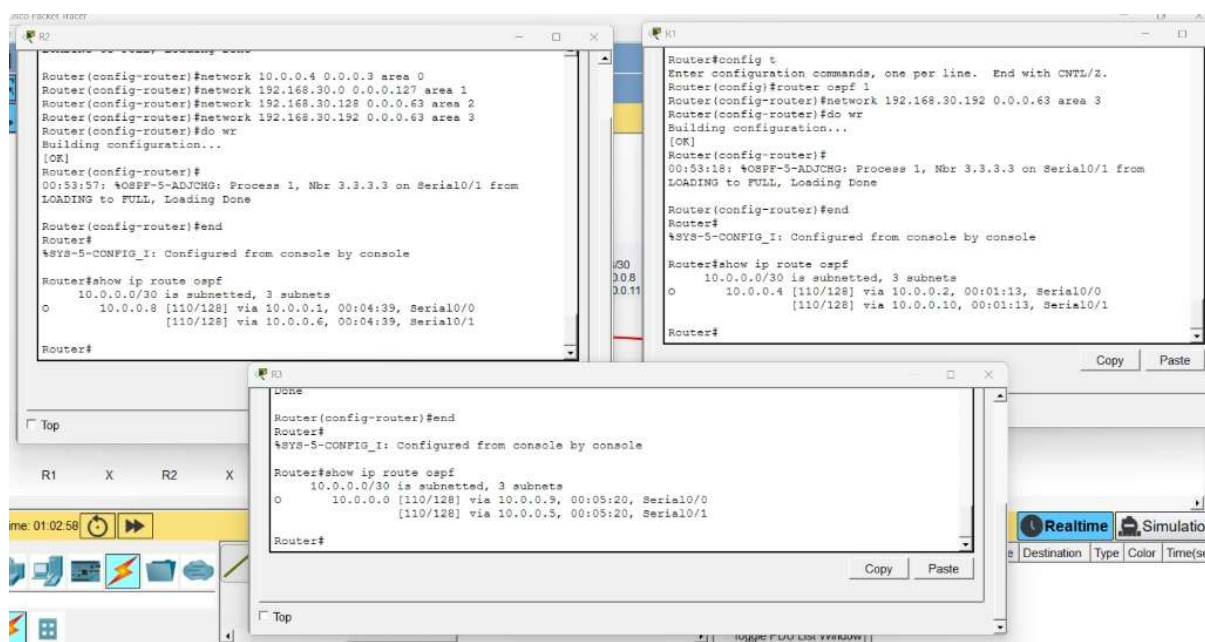


Figure 2 — Router CLI sessions immediately after OSPF configuration: the `%OSPF-5-ADJCHG` log messages confirm each neighbor reached the FULL state, and `show ip route ospf` lists the OSPF-learned /30 backbone routes on each router.

Task 4: Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router. Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.

The %OSPF-5-ADJCHG log lines seen in Figure 2 (for example, "Nbr 3.3.3.3 on Serial0/1 from LOADING to FULL") show the adjacency state machine completing on its own as each router exchanges Hello packets, elects a relationship over the point-to-point serial link, and synchronizes its link-state database. The show ip ospf neighbor command was then run on every router to capture a static snapshot of each adjacency.

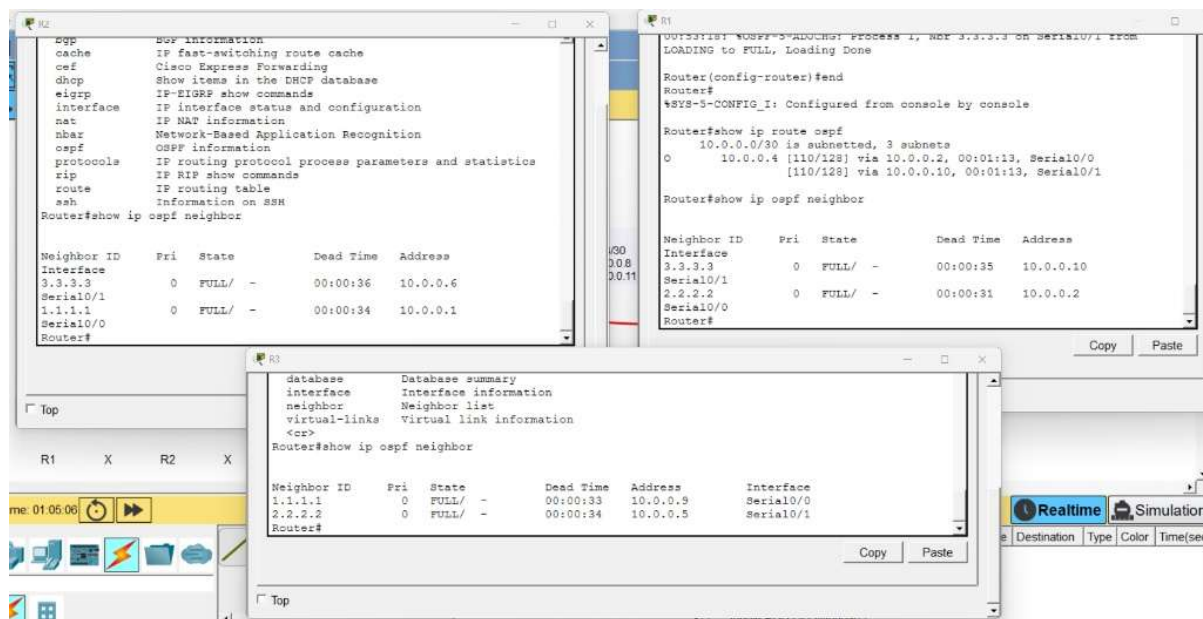


Figure 3 — show ip ospf neighbor output on R1, R2, and R3, each listing the other two routers' OSPF Router IDs in the FULL state.

Router	Neighbor ID	Neighbor Interface	State	Notes
R1	3.3.3.3	Serial0/1	FULL	Adjacency with R3 over 10.0.0.8/30
R1	2.2.2.2	Serial0/0	FULL	Adjacency with R2 over 10.0.0.0/30
R2	3.3.3.3	Serial0/1	FULL	Adjacency with R3 over 10.0.0.4/30
R2	1.1.1.1	Serial0/0	FULL	Adjacency with R1 over 10.0.0.0/30
R3	1.1.1.1	Serial0/0	FULL	Adjacency with R1 over 10.0.0.8/30
R3	2.2.2.2	Serial0/1	FULL	Adjacency with R2 over 10.0.0.4/30

Because these are point-to-point serial links rather than a shared broadcast segment (like Ethernet), OSPF does not elect a Designated Router (DR) or Backup Designated Router (BDR) here.

That DR/BDR election only applies on multi-access network types; on point-to-point links, every neighbor relationship is expected to reach plain FULL once the link-state databases are fully synchronized.

Task 5: Change the OSPF cost on one interface using the `ip ospf cost` command and verify how it affects the OSPF routing table. Hint: Use `show ip route ospf` to see route changes after modifying the cost

The command “Show ip ospf int se0/0” provides the cost for the route connected with that interface.

```
Router#show ip ospf interface Serial0/0

Serial0/0 is up, line protocol is up
 Internet address is 10.0.0.1/30, Area 0
 Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
 Transmit Delay is 1 sec, State POINT-TO-POINT,
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   Hello due in 00:00:03
 Index 1/1, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1 , Adjacent neighbor count is 1
   Adjacent with neighbor 2.2.2.2
 Suppress hello for 0 neighbor(s)
Router#show ip ospf interface Serial0/1

Serial0/1 is up, line protocol is up
 Internet address is 10.0.0.9/30, Area 0
 Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
 Transmit Delay is 1 sec, State POINT-TO-POINT,
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   Hello due in 00:00:05
 Index 2/2, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1 , Adjacent neighbor count is 1
   Adjacent with neighbor 3.3.3.3
 Suppress hello for 0 neighbor(s)
Router#
```

```
Router(config)#int se0/0
```

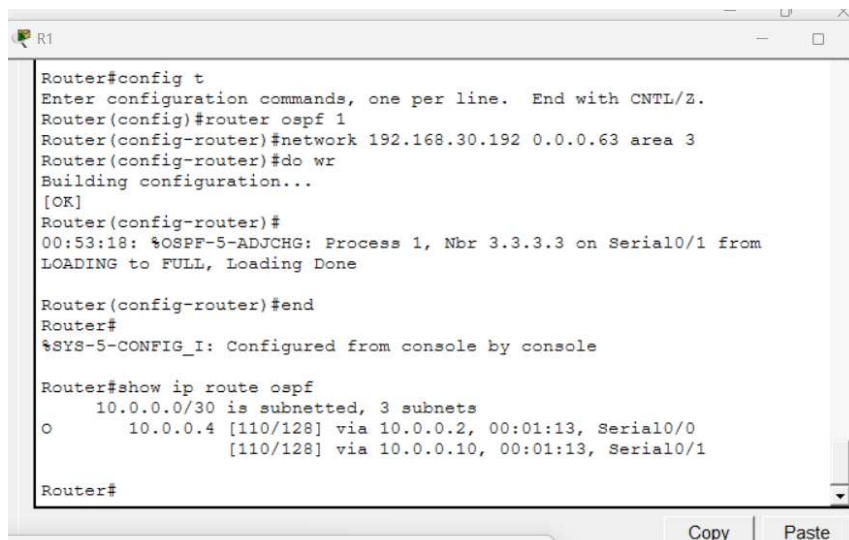
```
Router(config-if)#ip ospf cost ?
```

```
<1-65535> Cost
```

```
Router(config-if)#ip ospf cost 1000
```

```
Router(config-if)#end
```

This command will increase the bandwidth derived cost for serial 0/0.

A screenshot of a Cisco IOS command-line interface window titled 'R1'. The window shows a series of commands and their outputs. The commands entered are: 'Router#config t', 'Router(config)#router ospf 1', 'Router(config-router)#network 192.168.30.192 0.0.0.63 area 3', 'Router(config-router)#do wr', and 'Router(config-router)#end'. The outputs include 'Building configuration...', '[OK]', '00:53:18: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/1 from LOADING to FULL, Loading Done', and '%SYS-5-CONFIG_I: Configured from console by console'. Finally, the command 'Router#show ip route ospf' is entered, and the output shows the OSPF route table for area 3, indicating that the 10.0.0.0/30 network is subnetted into three subnets: 10.0.0.4 [110/128] via 10.0.0.2, 10.0.0.10 [110/128] via 10.0.0.10, and 10.0.0.16 [110/128] via 10.0.0.16. The window has a 'Copy' button and a 'Paste' button at the bottom right.

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.30.192 0.0.0.63 area 3
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:53:18: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/1 from
LOADING to FULL, Loading Done

Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route ospf
      10.0.0.0/30 is subnetted, 3 subnets
O       10.0.0.4 [110/128] via 10.0.0.2, 00:01:13, Serial0/0
          [110/128] via 10.0.0.10, 00:01:13, Serial0/1

Router#
```

The screenshot shows the cost value before the Cost Change

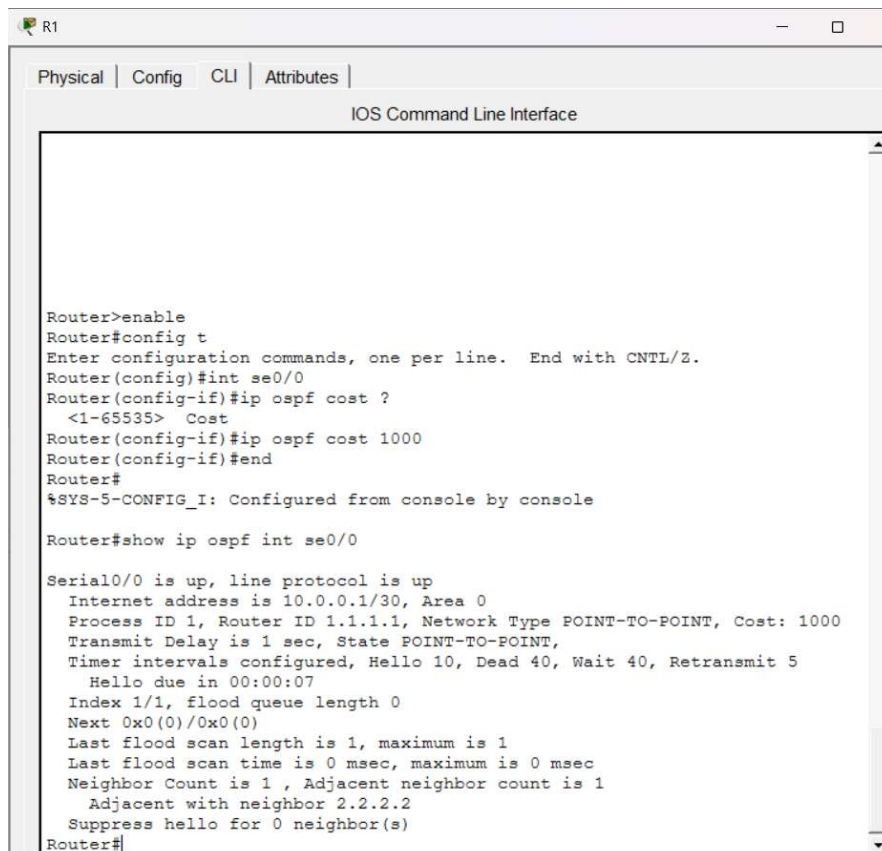
Originally, both serial interfaces (Serial0/0 and Serial0/1) had a default cost of 64.

- Because both paths to reach the remote 10.0.0.4/30 subnet had an equal cumulative cost ($64 + 64 = 128$), R1 performed Equal-Cost Multi-Path (ECMP) load balancing:

After Executing ip ospf cost 1000 on Serial0/0

By increasing the cost of Serial0/0 to 1000, the total metric for each path to destination 10.0.0.4/30 changed:

- Path via Serial0/0 (Direct link to R2): $1000 + 64 = 1064$
- Path via Serial0/1 (Indirect link via R3): $64 + 64 = 128$



```

R1
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se0/0
Router(config-if)#ip ospf cost ?
<1-65535> Cost
Router(config-if)#ip ospf cost 1000
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf int se0/0

Serial0/0 is up, line protocol is up
 Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 1000
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Router#

```

Routing Table Impact (show ip route ospf)

OSPF always selects the path with the lowest total cumulative metric.

As seen in your CLI output, R1 removed Serial0/0 from its routing table for that network and now routes all traffic to 10.0.0.4/30 exclusively through Serial0/1:

R1

Physical
Config
CLI
Attributes

IOS Command Line Interface

Link ID	ADV Router	Age	Seq#	Checksum
10.0.0.8	1.1.1.1	720	0x8000000b	0x00f40d
10.0.0.4	1.1.1.1	720	0x8000000c	0x009d27
10.0.0.0	1.1.1.1	314	0x8000000d	0x00f269

```

Router#
Router#show ip ospf ?
  <1-65535>      Process ID number
  border-routers Border and Boundary Router Information
  database       Database summary
  interface      Interface information
  neighbor       Neighbor list
  virtual-links  Virtual link information
  <cr>
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          0     FULL/ -         00:00:32    10.0.0.10      Serial0/1
2.2.2.2          0     FULL/ -         00:00:38    10.0.0.2       Serial0/0
Router#show ip route ?
  WORD           Network to display information about or hostname
  bgp            Border Gateway Protocol (BGP)
  connected      Connected
  eigrp          Enhanced Interior Routing Protocol (EIGRP)
  ospf           Open Shortest Path First (OSPF)
  rip            Routing Information Protocol (RIP)
  static         Static routes
  summary        Summary of all routes
  |             Output Modifiers
  <cr>
Router#show ip route ospf
      10.0.0.0/30 is subnetted, 3 subnets
O       10.0.0.4 [110/128] via 10.0.0.10, 00:07:47, Serial0/1
Router#

```

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